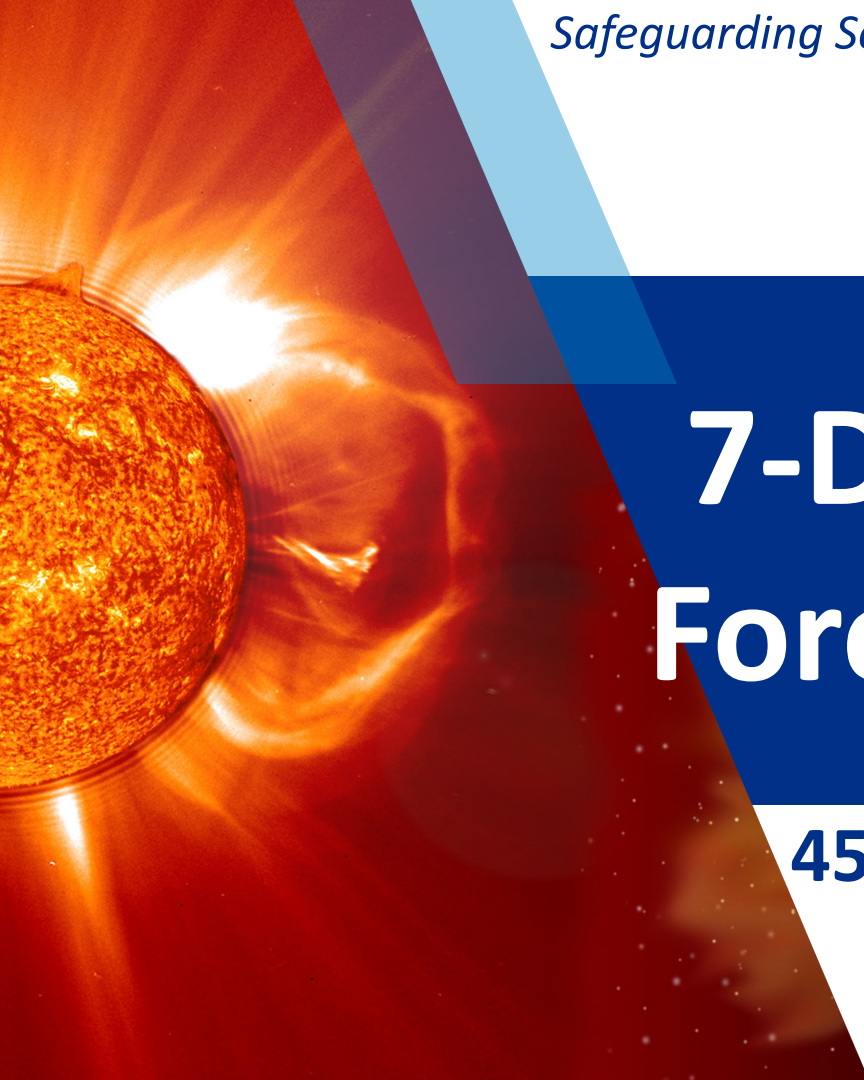




7-Day Kp and F10.7 Forecast Verification

45-Day Product User Engagement
January 21, 2026



Overview

- Introduction
- Data Summary
- F10.7 Analysis and Results
- Kp Analysis and Results
- Q&A



Introduction: Ap vs. Kp

Ap

- Linear (0-400)
- Units: nT



Kp

- Quasi-logarithmic (0-9)
- Unitless

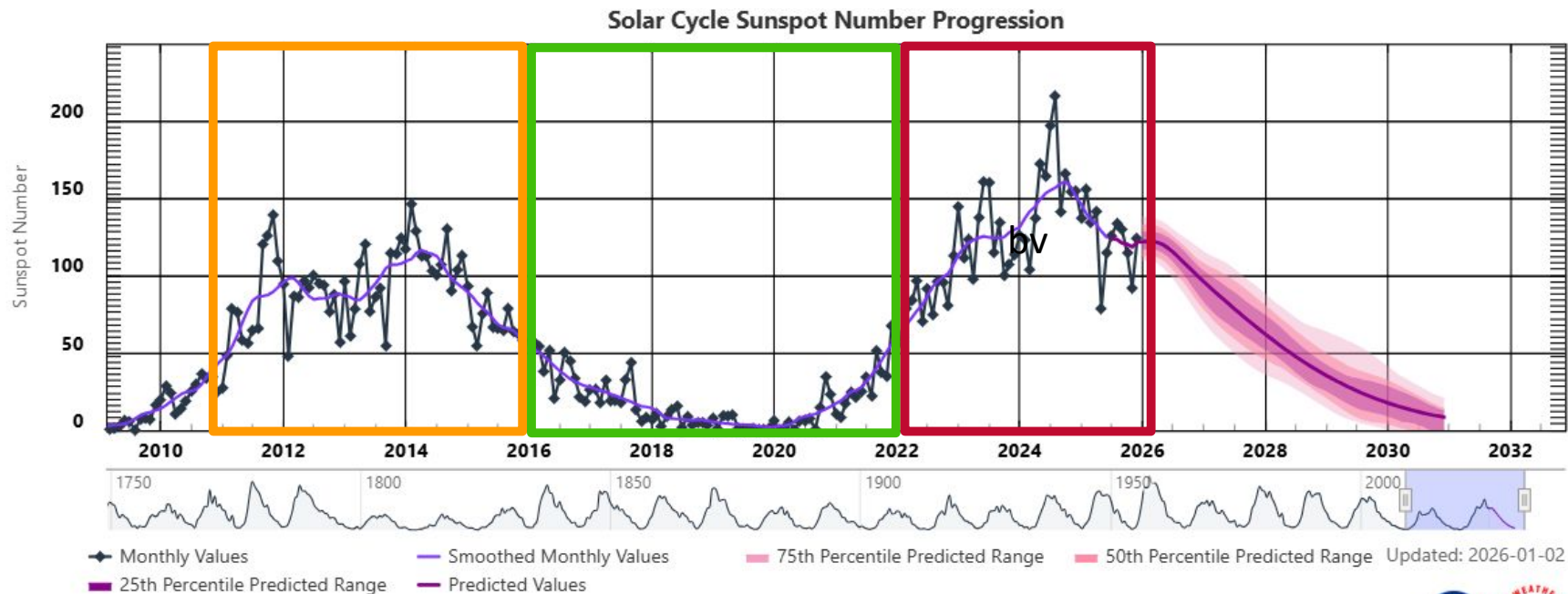
Kp = 0o 0+ 1- 1o 1+ 2- 2o 2+ 3- 3o 3+ 4- 4o 4+ 5- 5o 5+ 6- 6o 6+ 7- 7o 7+ 8- 8o 8+ 9- 9o

ap = 0 2 3 4 5 6 7 9 12 15 18 22 27 32 39 48 56 67 80 94 111 132 154 179 207 236 300 400



Introduction: Solar Cycle

- 2011-2016, 2016-2022, 2022-2016



<https://www.spaceweather.gov/products/solar-cycle-progression>



Data Summary: Data Retrieval

- Observational data from GFZ-Potsdam (final values)
 - https://www-app3.gfz-potsdam.de/kp_index/Kp_ap_Ap_SN_F107_since_1932.txt
- Forecast data from SWPC database (based on real-time values)
 - 7-day Kp forecasts (from November 2011)
 - 7-day F10.7 forecasts (from January 2011)
- Kp data pairs:
 - Total: 36,078 data pairs
 - 2011-2016: 10,437 data pairs
 - 2016-2022: 15,337 data pairs
 - 2022-2026: 10,304 data pairs
- F10.7 data pairs:
 - Total: 38,241 data pairs
 - 2011-2016: 12,689 data pairs
 - 2016-2022: 15,289 data pairs
 - 2022-2026: 10,263 data pairs

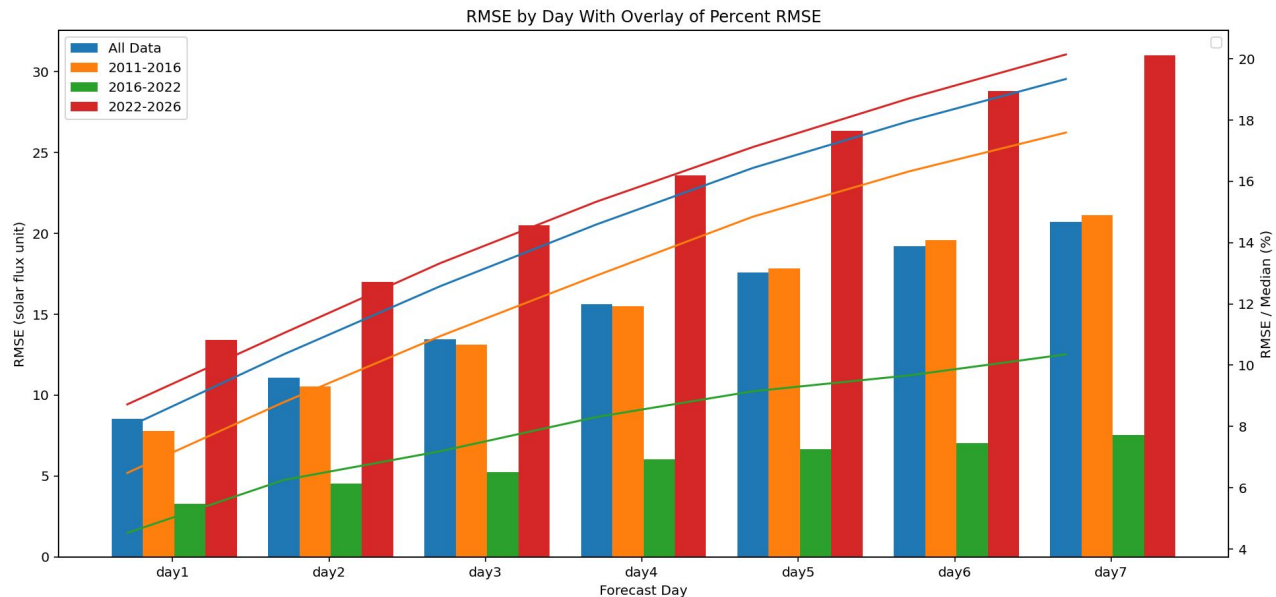




F10.7



F10.7 Results



Above: Lines → RMSE / Median & Bars → RMSE

- **Root Mean Square Error (RMSE):** square root of MSE, same units as values, sensitive to outliers like MSE
 - Lower is better!
- Low to moderate solar cycles have less error
- Error appears to increase linearly over 7 days
- Percent error ranges from ~10% (day 1) to ~20% (day 7)

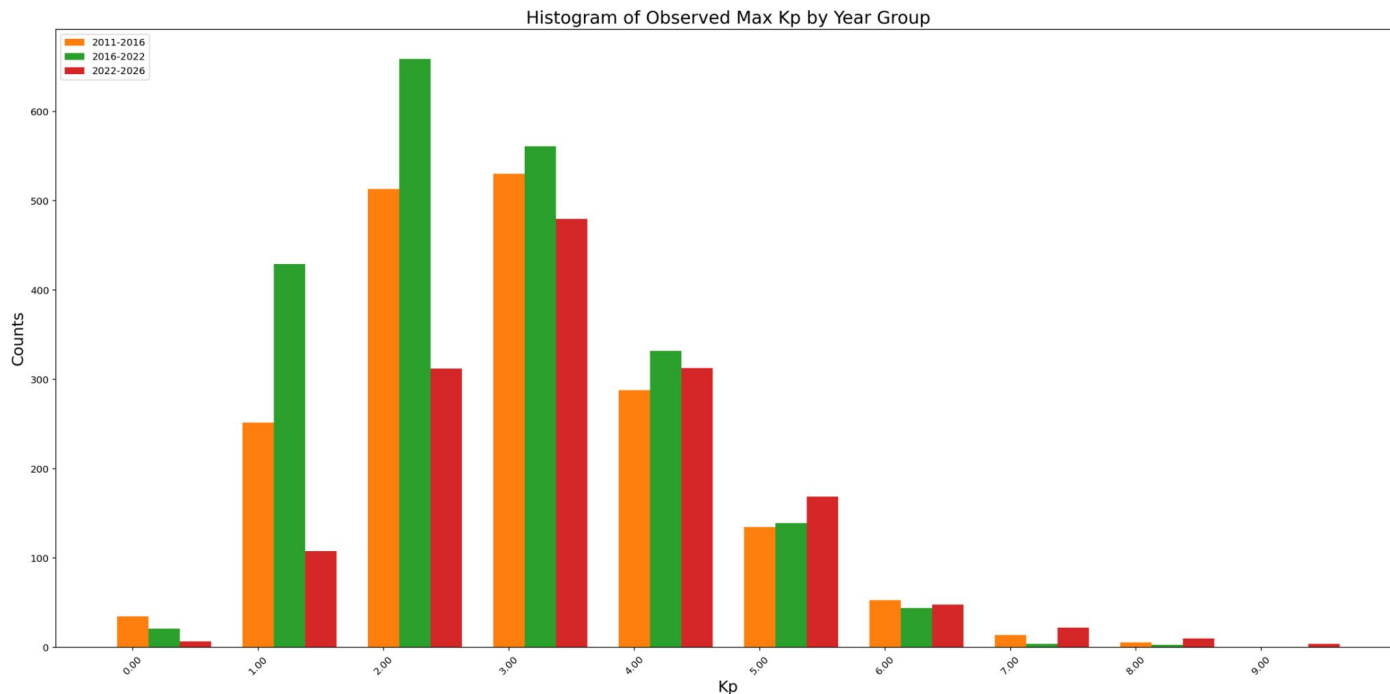




Kp



Kp Distribution - Observed Max



- 88.1% of observations < G1
- 98.8% of observations < G3
- Distribution is not Gaussian (normal)

Kp Strategy

- Two strategies: $\geq G1$ ($Kp \geq 4.667$) and $< G1$ ($Kp < 4.667$)
- Choose max Kp for each 24hr period and compare to each other and strategy threshold
- Example below:
 - Row 1: incorrect for both strategies
 - Row 2: Correct hit for $\geq G1$ strategy
 - Row 3: Correct hit for $< G1$ strategy

Obs Data

3hr 6hr 9hr 12hr 15hr 18hr 21hr 24hr

1	1	1	2	1	1	3	1
3	7	2	2	3	2	1	1
2	2	2	2	2	2	2	2



Forecast Data

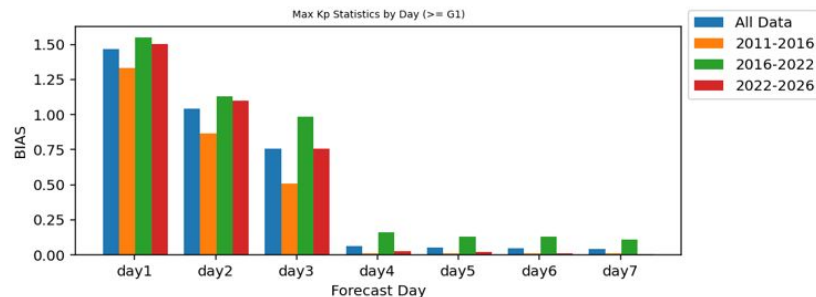
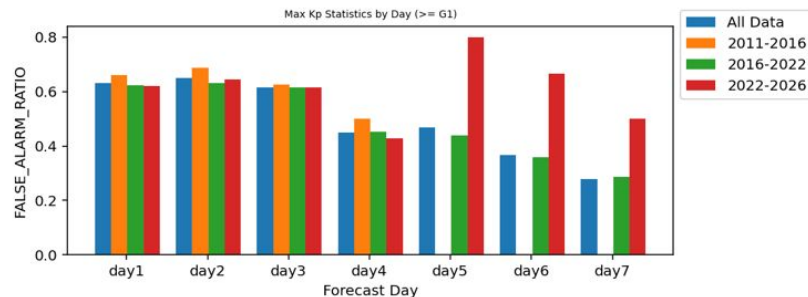
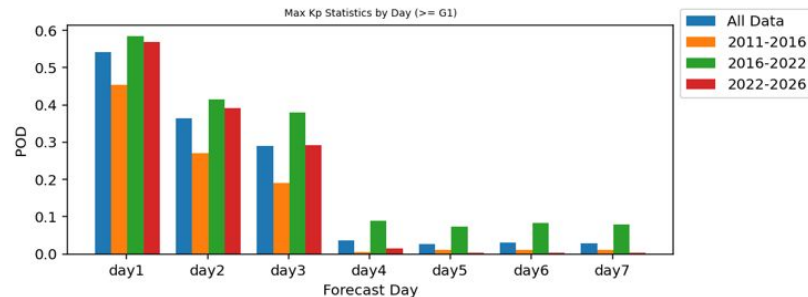
3hr 6hr 9hr 12hr 15hr 18hr 21hr 24hr

5	1	2	2	1	1	2	1
3	6	1	2	1	2	3	1
2	1	2	1	2	1	2	1



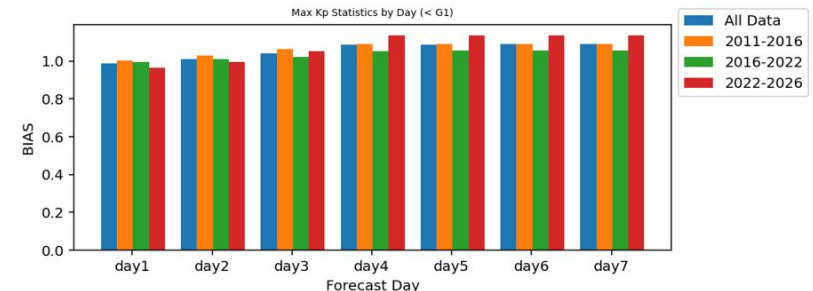
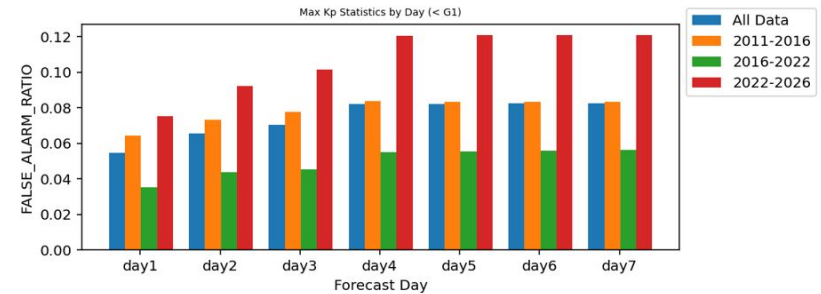
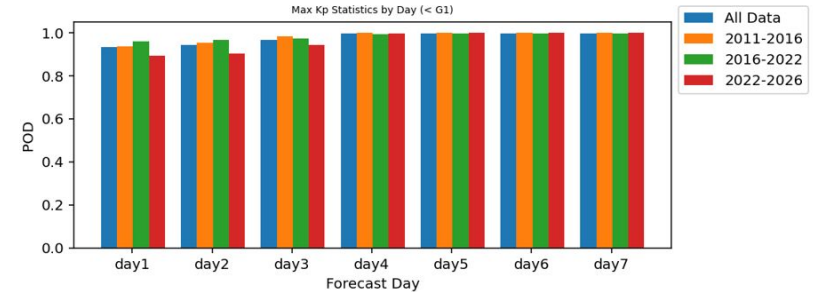
Kp Results: $\geq G1$ ($Kp \geq 4.67$)

- **Probability of Detection (POD):** Fraction of 'yes' observations forecast correctly, perfect = 1
- **False Alarm Ratio (FAR):** Fraction of 'yes' forecasts that were incorrect, perfect = 0
- **Bias Frequency:** Compares 'yes' forecast frequency to 'yes' observation frequency, perfect = 1



Kp Results: < G1 (Kp < 4.68)

- **Probability of Detection (POD):** Fraction of 'yes' observations forecast correctly , perfect = 1
- **False Alarm Ratio (FAR):** Fraction of 'yes' forecasts that were incorrect, perfect = 0
- **Bias Frequency:** Compares 'yes' forecast frequency to 'yes' observation frequency, perfect = 1



Kp Results Summary

- $\geq G1$
 - Large performance drop after day 3
- $< G1$
 - Does much better job discriminating between 'yes' and 'no' forecasts
 - Bias ~ 1 (i.e. distribution of forecasts closely matches observations)
- In general:
 - Solar cycle and activity level do not significantly impact the overall forecasting skill



Q & A





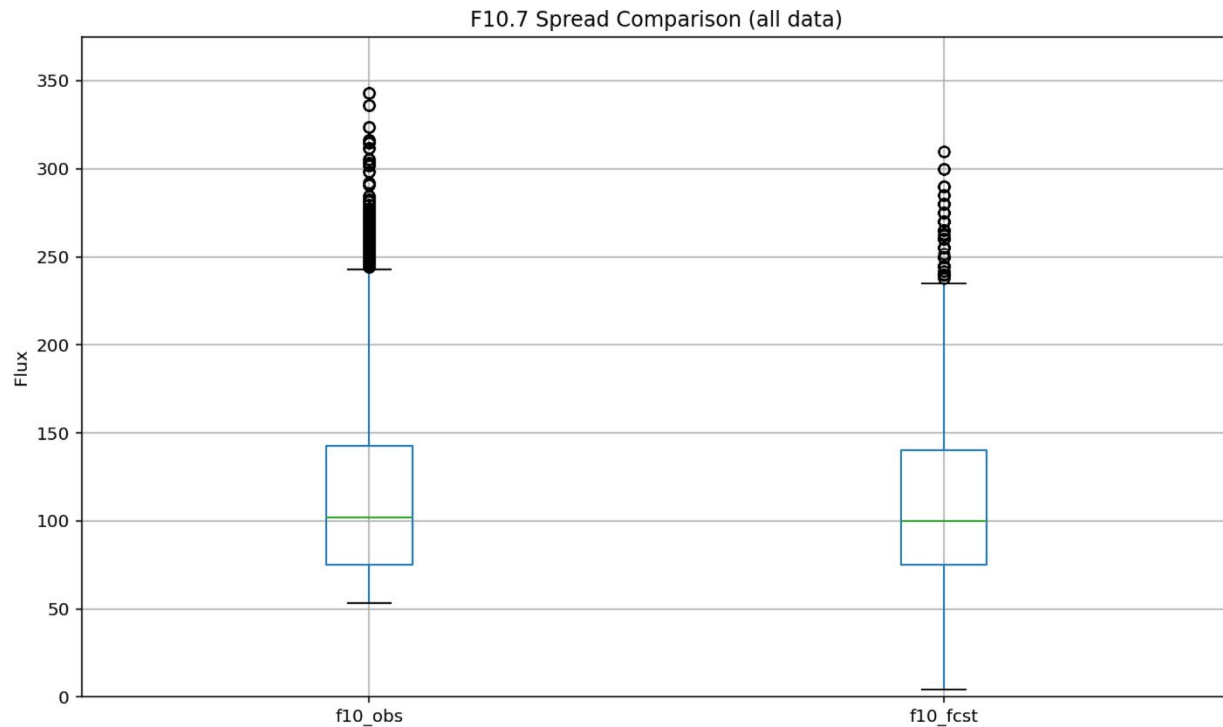
F10.7 Metrics

- **Mean Square Error (MSE):** average of squared differences between forecasted and observed values, sensitive to outliers
- **Mean Absolute Error (MAE):** average of magnitude of differences between forecasted and observed values, robust to outliers
- **Root Mean Square Error (RMSE):** square root of MSE, same units as values, sensitive to outliers like MSE
- **Pearson correlation coefficient:** strength of linear relationship between two variables
 - Assumes normality

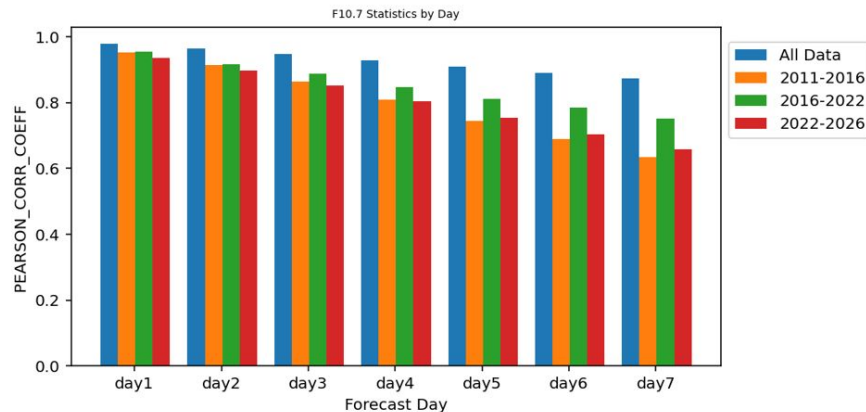
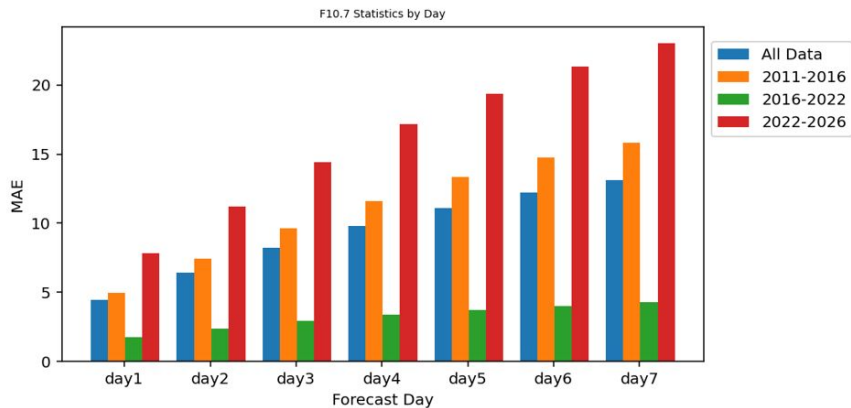
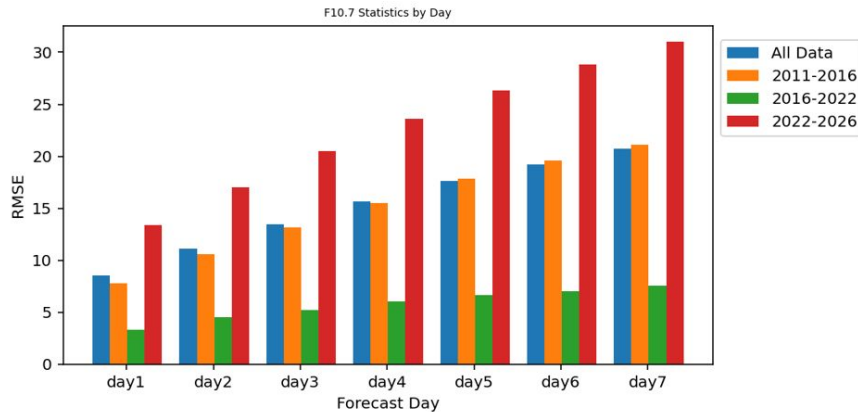
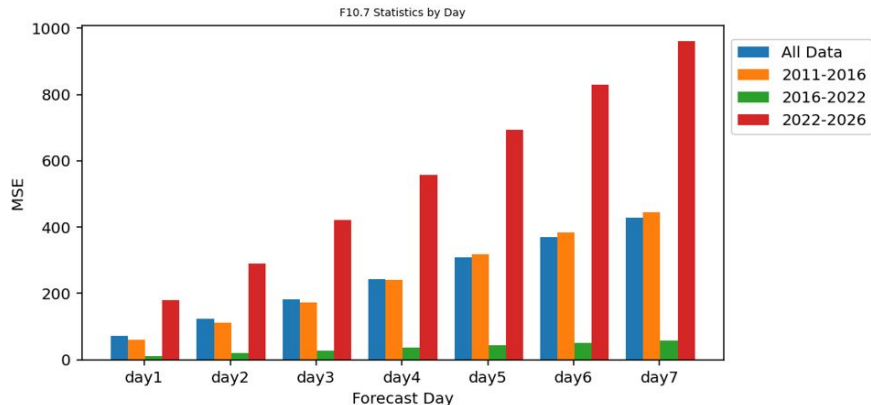


F10.7 Results

- Fairly similar spreads
- Nearly equal medians
- Lower first quartile limit for the forecasts



F10.7 Results





Kp Metrics

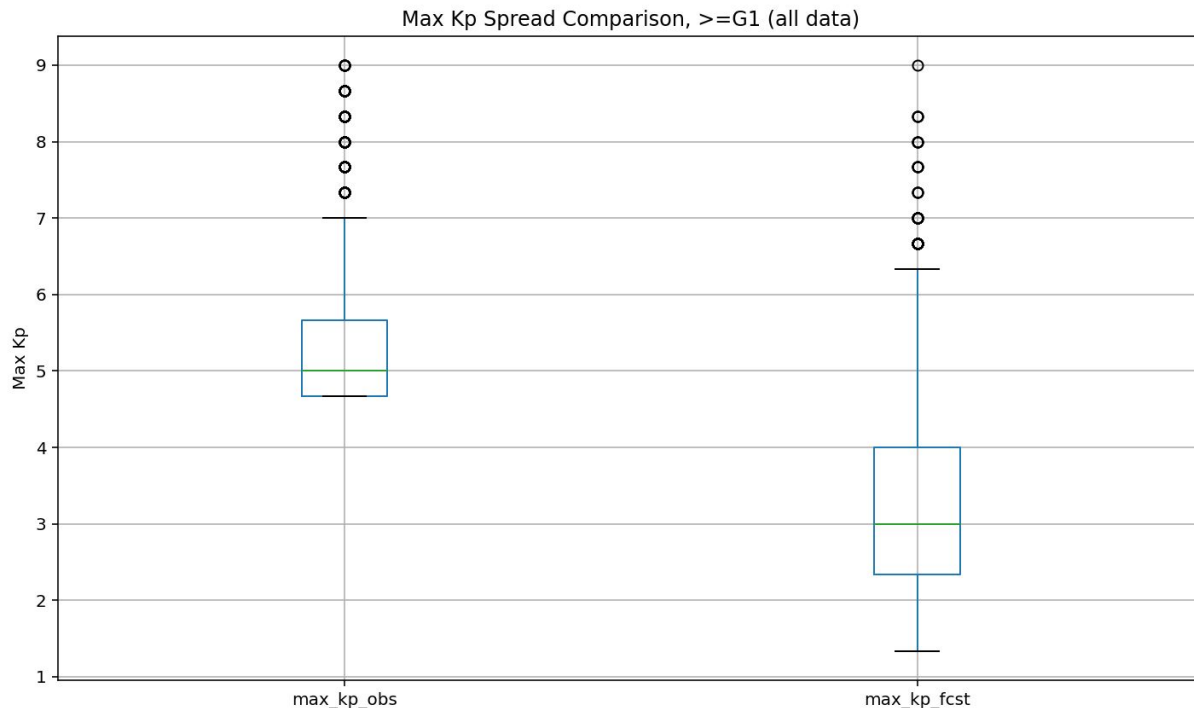
- **Bias Frequency:** Compares 'yes' forecast frequency to 'yes' observation frequency, perfect = 1
- **Probability of Detection (POD):** Fraction of 'yes' forecast correctly , perfect = 1
- **False Alarm Ratio (FAR):** Fraction of 'yes' forecast incorrectly, perfect = 0
- **Success Ratio:** Likelihood of event given a 'yes' forecast, perfect = 1
- **True Skill Statistic (TSS):** How well does forecast system discriminate between 'yes' and 'no', perfect = 1, no skill = 0
- **Heidke Skill Score (HSS):** Accuracy of forecast relative to random chance, perfect = 1, no skill = 0

https://www.cawcr.gov.au/projects/verification/#Methods_for_dichotomous_forecasts

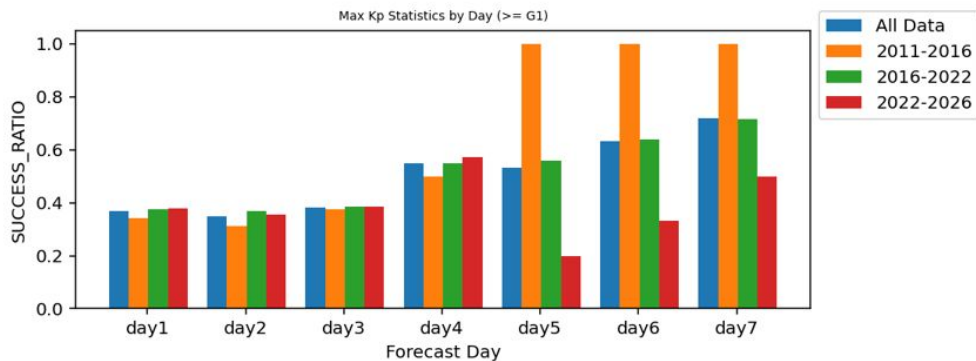
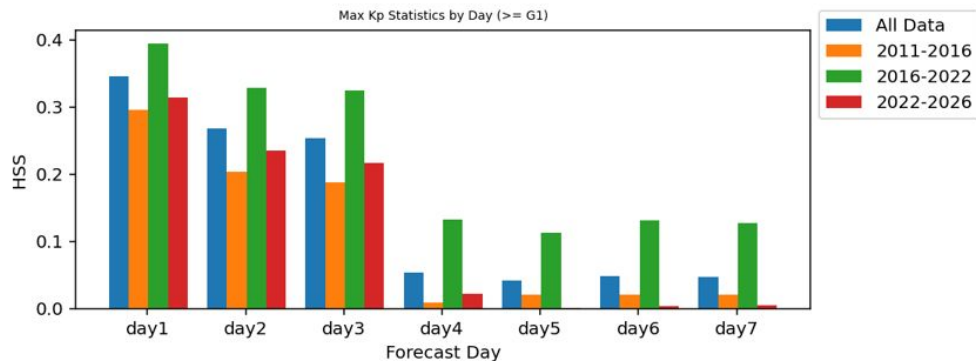
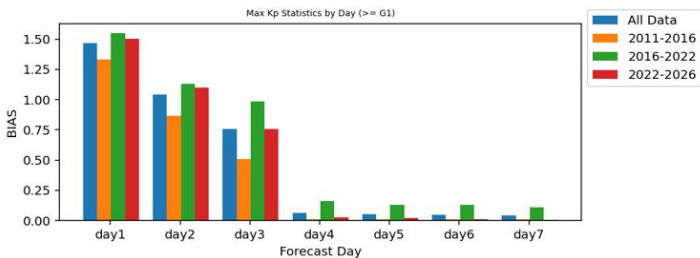
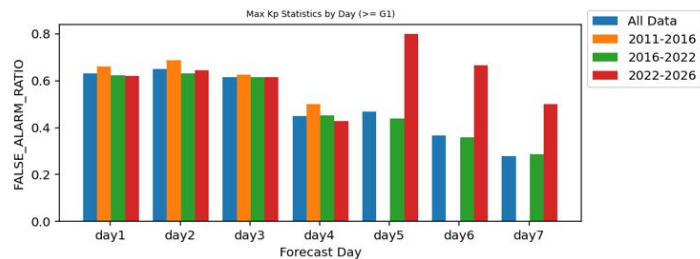
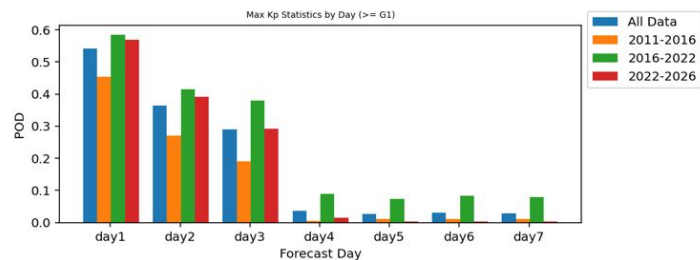


Kp Results: $\geq G1$ ($Kp \geq 4.67$)

- Data: If observed Max Kp ≥ 4.67 , the data pair was kept
- Vastly different spreads and medians

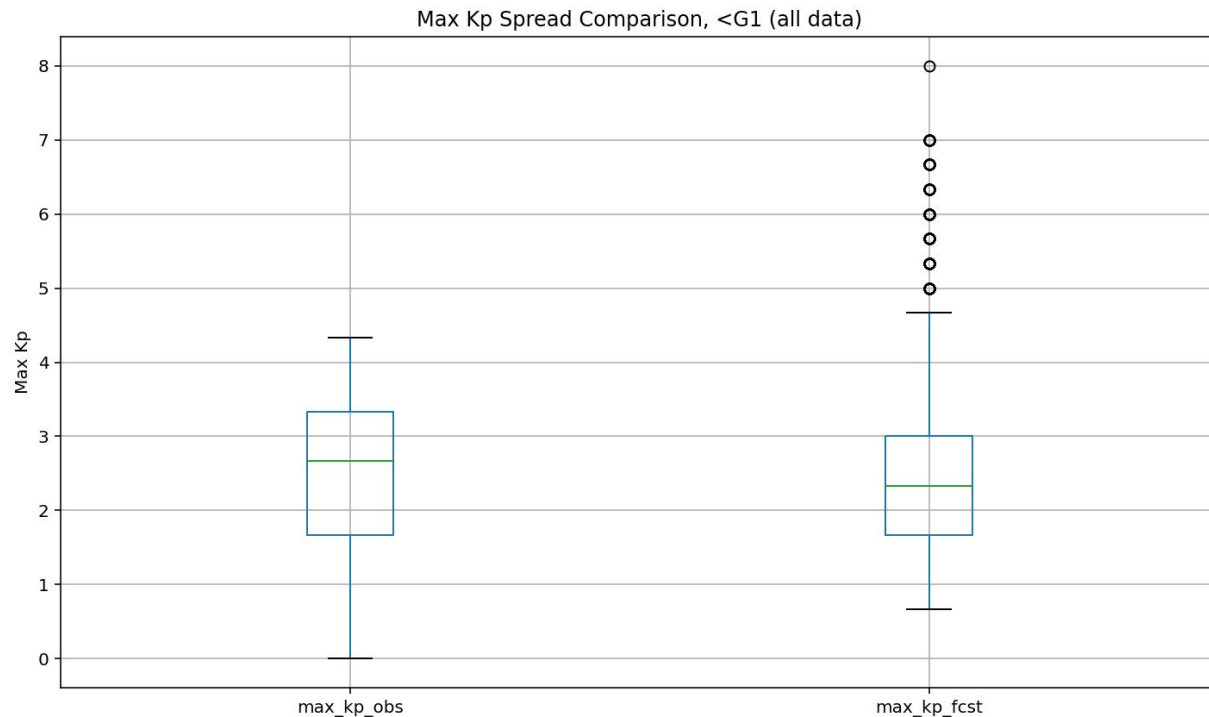


Kp Results: $\geq G1$ ($Kp \geq 4.67$)



Kp Results: < G1 (Kp < 4.67)

- Data: If observed Max Kp < 4.67, the data pair was kept
- Very similar medians



Kp Results: < G1 (Kp < 4.67)

